

SkinSync

Project Team

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Students' Submission:

This includes the title of the report and its occasion.

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Authors' Declaration

This states Authors' declaration that the work presented in the report is their own, and has not been submitted/presented previously to any other institution or organization.

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Introduction

In recent years, the demand for personalized solutions have been growing at a substantial rate, especially in industries focusing on healthcare and wellness. One major area gaining popularity is skincare, where individualized approaches are extremely important due to the complex interplay of factors like skin type, diet, and allergies. This project, **SkinSync**, aims to introduce an AI-driven skincare chatbot that not only provides tailored skincare advice but also could be utilised to incorporate dietary and other factors to offer a clean approach to skin health.

We at SkinSync will be leveraging many different technologies such as natural language processing (NLP), machine learning (ML), and image recognition, It is of the utmost importance that we deliver highly personalized recommendations to users, which could help them in maintaining optimal skin health and preventing any potential issues in the mere future. Unlike generic skincare advice, this chatbot will be analysing a user's data in real-time, as well as adapting its recommendations as their needs and other aspects evolve. By considering comprehensive factors such as lifestyle, nutrition, and potential allergens, this project intends to revolutionize the skincare industry with scientific, user-specific insights.

Abstract

SkinSync focuses on the development of an AI-powered chatbot that could be utilized to offer personalized skincare recommendations based on a person unique feature. By analyzing user-provided data such as skin type, dietary habits, and uploading images of an area a person wants information about, the chatbot delivers scientifically sound guidance that is customized for everyone who uses it. In addition to skincare, the chatbot will analyze nutritional deficiencies and lifestyle factors to provide a complete skin health assessment. Due to the advancements in ML and predictive analytics, the chatbot will continuously improve and adapt, offering more accurate recommendations each time an individual uses it. This approach will aim to bridge the gap between generic skincare solutions present on the internet and the personalized needs of each person, setting a new standard in the industry.

Executive Summary

Product developers have always focused on a one size fits all approach due to its feasibility, where skincare products and routines are marketed to the masses without considering individual needs. As a result, many different people have experienced severe frustration and unresolved skin issues just because there is a lack of personalized solution, We at SkinSync are here to deliver an AI-driven chatbot designed especially to address this gap by offering tailored made skincare advice based on each user's unique characteristics, including skin type, diet, and lifestyle. The project combines AI technologies such as ML, image recognition, and NLP to provide users with real-time, scientifically validated skincare recommendations.

By accomplishing this, individuals who will have access to our platform that will continuously learns and adapts to their needs, ultimately enhancing their overall skin health. Additionally, the application offers business potential by providing companies and dermatologists with a new, advanced tool for delivering personalized solutions to their clients.

Literature Review

Keeping Image classification as basis of our disease detection. We reviewed some reports that could be specific to our needs and be extremely useful during this process.

1. The paper "ETLoViT: an acne diagnose approach using vision-transformers and model ensembling" was authored by Krishna Veni Paluri and Ashish Gupta. It was published in **Engineering Research Express**, and it presents an excellent method for acne classification as it utilizes Vision Transformers for feature extraction, as well as coupled with deep learning architectures such as MobileNetV2, VGG16, and InceptionResNetV2 all which are mentioned approaches in the report. This approach led to a 96% accuracy in acne image classification, showcasing the superior performance compared to existing models present on other different platforms.

In their research, Paluri and Gupta utilized a dataset of only of 999 acne images, categorized these images into three different and unique levels, employed transfer learning was also utilized to enhance the model's performance. The ETLoViT model mentioned in the report as an innovative model demonstrated its ability to surpass state of the art skin lesion classification methods. As showcasing that by assembling multiple models, they achieved

improved classification accuracy, hence creating a brand new and promising avenue for automated acne diagnosis. The study gives emphasis on the significance of leveraging deep learning advancements to get more efficient, accurate medical image classification, especially when it comes to extremely complex conditions like acne.

2. In their chapter titled *Acne Detection Using Convolutional Neural Networks and Image Processing Technique*, the author Premanand Ghadekar and his colleagues presented an AI-based framework that was utilized to detect acne using CNNs. They explored many different applications of image processing techniques and used it to combine with CNN architectures to identify acne present in images efficiently and effectively. The methodology used by the authors included image pre-processing steps like noise reduction and segmentation, as well as enhancing the input data quality before utilizing it into the CNN model for training. The authors of this report highlighted the potential of their approach in automating dermatological diagnostics with higher accuracy compared to traditional visual inspection.

This work has significantly contributed to the growing research that applies deep learning to dermatology, particularly in acne detection. The researchers have delved into the challenges posed by acne diagnosis as well as show that their CNN-based system can be used to effectively classify and detect acne specifically by leveraging advanced image processing techniques.

3. In the article *A Novel Approach for Acne Detection Using Machine Learning*, the authors of this paper have written that by leveraging machine learning techniques we can enhance acne detection accuracy. The report has put major emphasizes on the use of various algorithms, including decision trees as well as support vector machines, alongside feature extraction methods, to put acne images into different classes effectively. By improving diagnostic precision, the study aims to address the limitations of traditional methods, thus offering a more reliable solution for skincare professionals and patients alike.

These three research papers align with our goal of finding solutions relevant to our project. The data used in this research can also prove to be relevant to our work and can be utilized for training.

Research on existing Products

	OLAY	Spotscan	CureSkin
Features	Analyzes user selfies to determine "skin age" and suggests Olay products.	Evaluates acne severity from photos and recommends La Roche-Posay products.	Provides personalized skincare solutions and treatments for various skin issues.
Limitations	Limited to recommending only Olay products, restricting broader skincare advice.	Limited to recommending only La Roche-Posay products, restricting unbiased recommendations.	Restricted to India, limiting global reach and user impact.

Table 1

Project Vision Problem Statement

Problem:

The average human being often struggles to identify skincare routines that is suitable to their unique skin types, dietary habits, which could inevitably lead to ineffective treatments and try and testing approach leads to unnecessary expenses. Additionally, there is a gap in addressing men's skincare needs due to societal norms.

Impact:

Currently, skincare problems like acne and hyperpigmentation affect millions globally, leading to self-esteem issues as well as mental health challenges. According to studies, untreated skin conditions often result in long term physical and psychological impacts, such as depression, anxiety, and could also cause social withdrawal. Moreover, the skincare industries has a certain bias towards certain demographics leaving many undervalued. By addressing these issues with our solutions, *SkinSync* we can empower users to take control of their skin health, overall improving their well-being and confidence (Jones & Chan, 2023).

Objectives

- **Objective 1:** Develop an AI model that is capable of analysing skin images and detect conditions such as acne, hyperpigmentation, or other skin conditions.
- **Objective 2:** Provide personalized dietary recommendations to address any skin deficiencies present as well as any related health concerns.
- **Objective 3:** Implement predictive analytics that could track user skin health over time and offer future product suggestions based on changes in skin condition overtime.
- **Objective 4:** Make sure the chatbot is accessible across many different income levels, hence being an affordable product options and reduce the stigma around men's skincare practices.

Scope

- We are developing an AI-powered chatbot that will provide personalized skincare recommendations based on skin type, dietary habits, potential allergens.
- SkinSync will also let you incorporate image recognition for skin analysis.
- We will also provide nutrition deficiency detection and supplement recommendations.
- Integrate APIs into our system for skincare product recommendations.
- We will by addressing skincare concerns for all genders, focusing on all genders, often overlooked due to societal norms.
- Prioritizing accessibility for users that have limited skincare knowledge or have any financial constraints.

Constraints

- First there is limited dataset availability for niche skin types.
- Privacy concerns is one of the biggest constraints of any application especially with user health data.
- Dependency is present on accurate API data for product recommendations.

Stakeholders Description

3.6.1. Stakeholders Summary

Stakeholders	Description
Consumers	Individuals seeking personalized skincare advice and product recommendations.
Dermatologist	Professionals who may use the system to complement their practice and offer tailored advice to patients.
Skincare product companies	Businesses that could benefit from the system by having their products recommended to the appropriate target audience.
Health and wellness companies	Organizations interested in integrating the chatbot's capabilities into their services.
AI and ML Researchers	Academics and professionals focused on advancing AI applications in healthcare.
Software Developers	The team responsible for building and maintaining the AI-powered chatbot.
API Providers	Third-party services that supply data for skincare products, allergens, and other relevant information.

Table 2

Key High-Level Goals and Problems of Stakeholders:

- *Users: they want accurate, affordable, as well as accessible skincare advice that is tailored to their specific needs.*

Problems: Main Concern about data privacy as different techniques could cause different issues, there is also lack of personalized skincare products, misinformation.

- *Healthcare Professionals: They use AI to streamline diagnosis as well as initial consultation.*

Problems: there is a fear of AI misdiagnosis as could cause errors, due to which there is less trust in automated systems.

- *Skincare Companies: there is an increase product visibility and engagement through personalized recommendations as their product could be advertised to consumers*

Problems: however main concern could be the overcoming market saturation and competition.

- *Developers: Ensure the chatbot remains updated with the latest skincare science.*

Problems: Managing AI training data and system scalability.

- *Regulatory Bodies: Safeguard user privacy and ensure the system aligns with healthcare regulations.*

Problems: Ensuring compliance across different regions.

Software Requirement Specifications

List of Features

1. User Interface

- An interactive and user-friendly interface.
- A chat window for users to communicate.
- Image uploads for facial detection and analysis.
- User authentication and profile management.

2. Chatbot

- Primary tool for interacting with the AI dermatologist.
- Handles inquiries and ensures smooth conversational flow.
- Collaborates with diagnostic and recommendation systems to provide accurate and useful responses.
- Delivers natural and informative user experiences.

3. Image Processing & Facial Detection

- Detects and analyzes the user-specified skin region in uploaded images.
- Prepares images by filtering, normalizing, and extracting features for analysis.

4. Disease Diagnosis

- Utilizes advanced machine learning to analyze processed images and diagnose skin conditions, such as acne.
- Classifies skin condition characteristics and provides a diagnostic accuracy score.

5. Product Recommendation

- Recommends skincare products based on diagnosed skin conditions.
- Leverages a product database to suggest personalized options that match the user's skin type and needs.

6. Knowledge Base & NLP

- Understands user input using Natural Language Processing (NLP).
- Refers to a knowledge base to deliver accurate and relevant responses to user queries.

7. Backend & Database Management

- Manages databases, processing, and retrieval operations.
- Handles user data, images, diagnosis results, and product information.
- Ensures secure and efficient data management via APIs and databases.

Functional Requirements

1. Registration:

- A. User shall be able to create an account if there is no existing account associated with the user, email-id.
- B. Users shall be able to enter account details. All these are required fields and account creation cannot proceed unless these are filled.
- C. User registers after this information is validated and the user agrees with the terms and conditions.

2. Login Image Analysis for Skin Conditions:

- A. The chatbot can accept and analyze skin condition images using CNNs and image recognition.
- B. Dependency: This feature is essential as the Personalized Product Recommendations and Medical Advice rely on accurate diagnosis from this analysis.

3. Personalized Product Recommendations:

- A. The chatbot suggests skincare products based on the diagnosis.
- B. Dependency: Depends on the output of the Image Analysis for Skin Conditions to make accurate product suggestions.

4. Dietary Recommendations:

- A. Provides dietary advice based on skin condition.
- B. Dependency: It is independent but can benefit from data gathered from the Personal Health Questionnaire.

5. Medical Advice and Referrals:

- A. Recommends seeing a dermatologist for severe cases.
- B. Dependency: Relies on the diagnosis from Image Analysis to determine if a referral is necessary.

6. User Data and History Management:

- A. Tracks user interaction history for future reference.
- B. Dependency: None directly, but improves the overall user experience by retaining historical data for future Image Analysis and Recommendations.

7. Multilingual Support:

- A. Supports multiple languages to enhance accessibility.
- B. Dependency: Independent of the other FRs but must be integrated into each feature to ensure functionality across different languages.

8. Personal Health Questionnaire:

- A. Asks users about lifestyle factors for better accuracy.
- B. Dependency: Feeds data into Dietary Recommendations and enhances the overall diagnosis accuracy for Personalized Product Recommendations.

Non - Functional Requirements

1. Performance Requirements:

- A. **Response Time:** The system should respond to user interactions, such as generating recommendations or detection, without extra delay.
- B. **Scalability:** The application should be able to handle a considerable amount of users without significant performance degradation.
- C. **Availability:** The system should have minimized disruption and be accessible when needed. The system should be reliable.

2. Security Requirements:

- A. The users must be authenticated before accessing the system.
- B. The system should employ control mechanisms to restrict user access to only the necessary resources and features.
- C. The system shall perform all input checks before allowing the user to register.

3. Compatibility:

- A. The system shall be able to run seamlessly on current day browsers. For example Chrome, Mozilla Firefox.
- B. The system shall be able to run on various devices For example, Tablet, Laptop and mobile phone.

4. Maintainability:

- A. The system shall be developed keeping in mind the extension if needed. Incorporation of new features should be easy to modify the existing features if needed.

5. Documentation:

- A. A comprehensive documentation shall be provided to the user for understanding the guidelines and responsible use of the system. This can include a user manual, video tutorial and frequently asked questions.

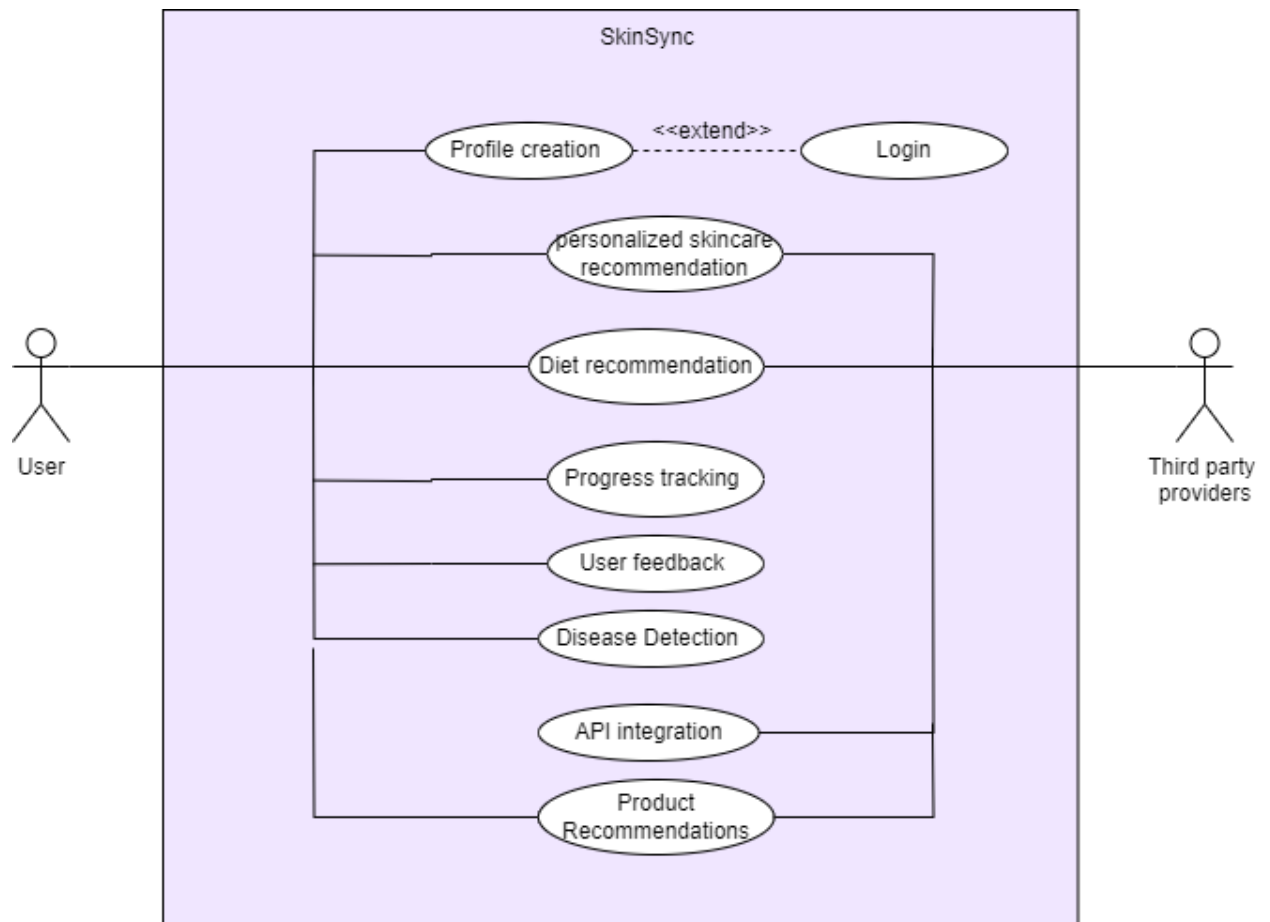
Iteration Plan

Iteration	Time frame	Module(s)
1	FYP 1 Mid Evaluation	UI, Disease diagnosis
2	FYP 1 Final Evaluation	Chatbot, backend
3	FYP 2 Mid Evaluation	Product Recommendation, Integration
4	FYP 2 Final Evaluation	Interface refinement, Optimization

Table 3

Iteration 1

Use Case Diagram



High level Use Case

Use Case ID	01
Use Case	Profile creation
Actor	User
Type	Primary
Description	Users will create an account to share information needed to create an account

Use Case ID	02
Use Case	Skincare recommendation
Actor	User, third party providers
Type	Primary

Description	The system provides users with customized routine suggestions based on their profile.
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Use Case ID	03
Use Case	Diet recommendation
Actor	User, third party providers
Type	Primary
Description	Third party would provide with dataset, Users will be given the correct recommendation according to their diagnosis

Use Case ID	04
Use Case	Progress tracking
Actor	User
Type	Primary
Description	The system keeps track of users' skincare routines, product usage, and overall progress, presenting it in a visual format.

Use Case ID	05
Use Case	User feedback
Actor	User, System
Type	Primary
Description	Users submit feedback on the effectiveness of the recommendations and products

Use Case ID	06
Use Case	Disease Detection
Actor	User, System
Type	Primary
Description	The system analyzes user data and predicts potential skin issues before they occur

Use Case ID	07
Use Case	API integration
Actor	Third party provider
Type	Primary
Description	The system integrates external data sources, such as allergen levels, and product databases, to enhance the accuracy of recommendations.

Use Case ID	08
Use Case	Product recommendation
Actor	User, Third party providers
Type	Primary
Description	The system provides immediate skincare product recommendations based on real-time data

Extended Use Cases

Use Case Name	Personalized Skincare Recommendations
Scope	Primary
Level	User level
Primary Actor	User (person seeking personalized skincare advice)
Stakeholders & Interests	• User: Wants tailored skincare recommendations • Dermatologist: Interested in reviewing or adjusting recommendations • System Provider: Ensures high user satisfaction • Third-Party Providers: Benefit from integrating their data into recommendations.
Preconditions	• User has completed their profile. • System can retrieve real-time data from third-party providers. • Access to skincare product databases.
Postcondition	• User receives and saves personalized skincare recommendations.

	Recommendations are stored in the user's history.												
Main Success Scenario	<table> <tr> <th>Actor Action</th><th>System Response</th></tr> <tr> <td>1. User navigates to the recommendation section</td><td></td></tr> <tr> <td></td><td>2. System retrieves user profile, dietary</td></tr> <tr> <td></td><td>3. analyses the information and generates personalized product and routine suggestions</td></tr> <tr> <td></td><td>4. System presents the recommendations</td></tr> <tr> <td>5. reviews and saves recommendations</td><td></td></tr> </table>	Actor Action	System Response	1. User navigates to the recommendation section			2. System retrieves user profile, dietary		3. analyses the information and generates personalized product and routine suggestions		4. System presents the recommendations	5. reviews and saves recommendations	
Actor Action	System Response												
1. User navigates to the recommendation section													
	2. System retrieves user profile, dietary												
	3. analyses the information and generates personalized product and routine suggestions												
	4. System presents the recommendations												
5. reviews and saves recommendations													
Alternative	A1: User has not completed profile – system prompts for missing data. A2: No real-time data available – system generates recommendations based on profile only												
Exceptions	E1: No internet connection – system informs user of limited recommendations. E2: Product database unavailable – system notifies user and suggests retrying later.												

Use Case Name	Profile Creation
Scope	Primary
Level	User level
Primary Actor	User
Stakeholders & Interests	User: Wants to create a personalized skincare profile. - System Provider: Gathers accurate data for better recommendations.
Preconditions	User has signed up on web page
Postcondition	User profile is created, stored in the system, and used for personalized recommendations.

Main Success Scenario		
	Actor Action	System Response
	1. selects “Create Profile” and inputs necessary details.	
		2. validates and saves the profile.
		3. confirms successful creation.
Alternative	A1: User fails to provide all required information – system prompts user to complete missing fields.	
Exceptions	E1: System fails to save data due to a technical issue – user is informed and asked to try again later.	

Use Case Name	Dietary and Nutritional Assessment
Scope	Primary
Level	User level
Primary Actor	User
Stakeholders & Interests	User: Wants to improve diet for better skin health. System Provider: Gathers accurate dietary data to enhance recommendations. Third-Party Providers: Provide nutritional data.
Preconditions	User has completed their profile and provided basic dietary information.
Postcondition	User receives dietary recommendations related to skin health.

Main Success Scenario	Actor Action	System Response
	User submits dietary details.	
		2. System analyzes data and accesses third-party sources for nutrition info.
		3. System provides dietary suggestions
Alternatives	A1: Incomplete dietary information – system prompts user for more data. A2: No third-party data – system makes suggestions based only on user input.	
Exceptions	E1: System fails to retrieve third-party data – user informed of limited suggestions.	

Use Case Name	Real-Time Product Recommendations
Scope	Primary
Level	User level
Primary Actor	User
Stakeholders & Interests	• User: Wants real-time product suggestions. • System Provider: Gathers real-time data to improve user experience. • Third-Party Providers: Provide real-time data
Preconditions	User has a completed profile and the system is connected to real-time third-party data.
Postcondition	User receives product suggestions based on current environmental

	conditions.	
Main Success Scenario		
	Actor Action	System Response
	1. requests real-time product suggestions.	
		2. System retrieves real-time data (allergens, etc.).
		3. System provides recommendations based on user profile and real-time data.
	4. User views and saves suggestions.	
Alternative	A1: No real-time data available – system provides recommendations based only on the user's profile.	
Exceptions	E1: System fails to retrieve real-time data – user is notified of limitations.	

Use Case Name	Predictive Skin Issue Detection
Scope	Primary
Level	User level
Primary Actor	User
Stakeholders & Interests	• User: Wants to prevent potential skin issues. • Dermatologist: Interested in early detection of issues. • System Provider: Ensures accurate predictions for user health
Preconditions	User profile and history are up-to-date. system is connected to third-party data for accurate prediction
Postcondition	User is informed of potential skin issues and recommended preventive actions.

Main Success Scenario		
	Actor Action	System Response
	1. uploads skin health.	
		2. System analyzes profile, history, and environmental data.
		3. System identifies potential issues and alerts the user.
	User receives preventive recommendations.	
Alternative	A1: No new issues detected – system informs user that no issues are predicted.	
Exceptions	E1: Predictive analysis fails – system notifies user to try again later.	

Use Case Name	User Feedback
Scope	Primary
Level	User level
Primary Actor	User
Stakeholders & Interests	User: Wants to provide feedback to improve the system. System Provider: Interested in gathering feedback to improve service quality.
Preconditions	User has interacted with the system and has feedback to share

Postcondition	Feedback is submitted and logged for review by the system provider.											
Main Success Scenario	<table><tr><th>Actor Action</th><th>System Response</th></tr><tr><td>1. selects feedback submission option</td><td></td></tr><tr><td>2. enters their feedback.</td><td></td></tr><tr><td></td><td>3. validates and submits feedback.</td></tr><tr><td></td><td>4. confirms successful submission</td></tr></table>		Actor Action	System Response	1. selects feedback submission option		2. enters their feedback.			3. validates and submits feedback.		4. confirms successful submission
Actor Action	System Response											
1. selects feedback submission option												
2. enters their feedback.												
	3. validates and submits feedback.											
	4. confirms successful submission											
Alternative	A1: Feedback is incomplete – system prompts user for more details.											

Use Case Name	Progress Tracking
Scope	Primary
Level	User level
Primary Actor	User
Stakeholders & Interests	- User: Wants to track their skin progress over time. - Dermatologist: Reviews progress for any necessary interventions. - System Provider: Ensures accurate tracking for user engagement.
Preconditions	The user is logged into his account.
Postcondition	User history and progress are tracked and viewable over time

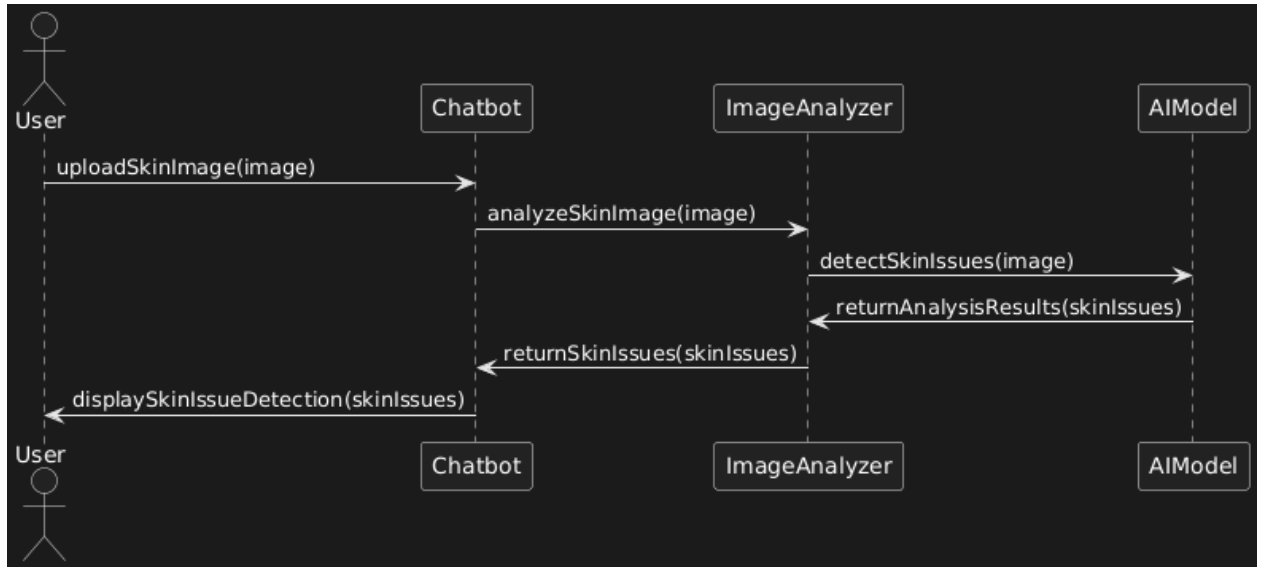
Main Success Scenario		
	Actor Action	System Response
	1. User views history and progress.	
		2. System displays tracked skin data over time.
	.3. User compares progress and views potential improvements.	
Alternative	A1: No significant progress – system informs user of lack of change	

Use Case Name	API Integration
Scope	Primary
Level	User level
Primary Actor	Third-Party Data Providers
Stakeholders & Interests	- System Provider: Needs reliable third-party data for real-time skincare recommendations. - Third-Party Providers: Interested in providing accurate data for integration.
Preconditions	API connections with third-party providers are established.
Postcondition	The system retrieves and integrates third-party data into skincare recommendations.

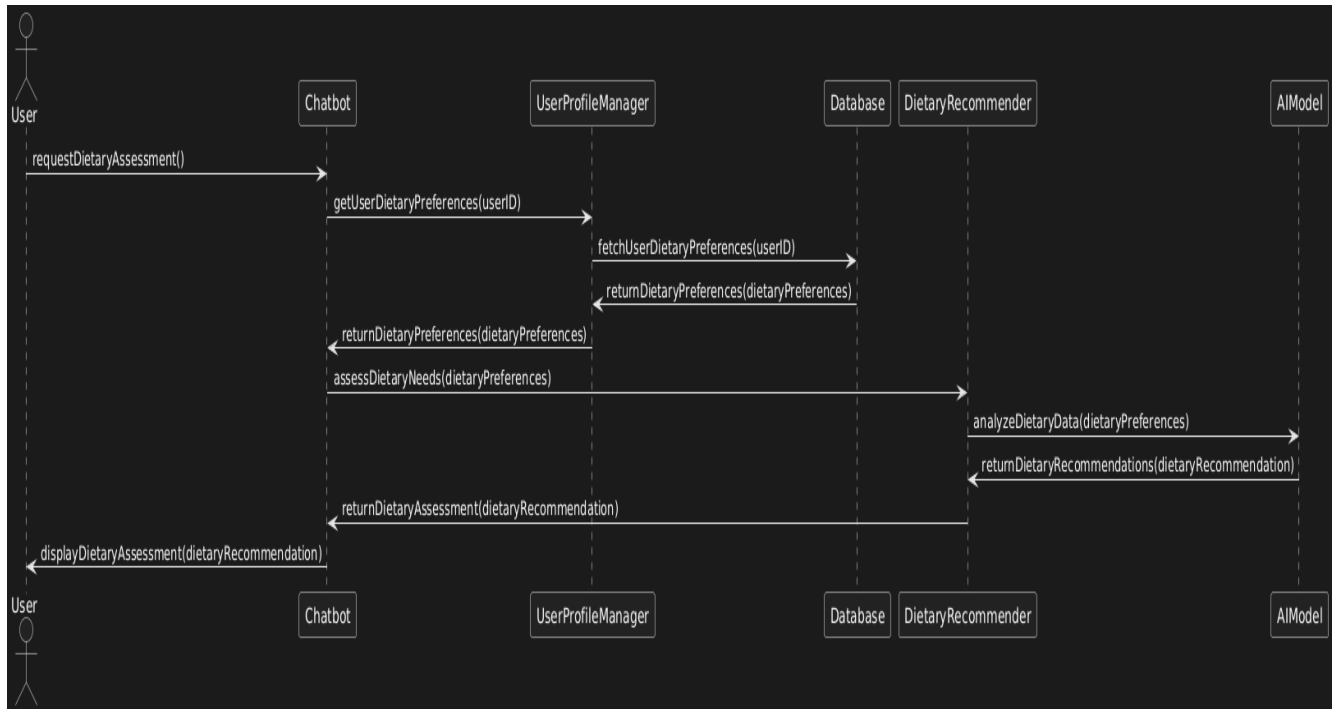
Main Success Scenario		
	Actor Action	System Response
		1. System connects to third-party APIs.
		2. system retrieves relevant data (weather, allergens, etc.)..
		3. System integrates the data into real-time recommendations
Alternative	A1: API connection fails – system reverts to basic recommendations based on user profile only.	

System Sequence Diagrams

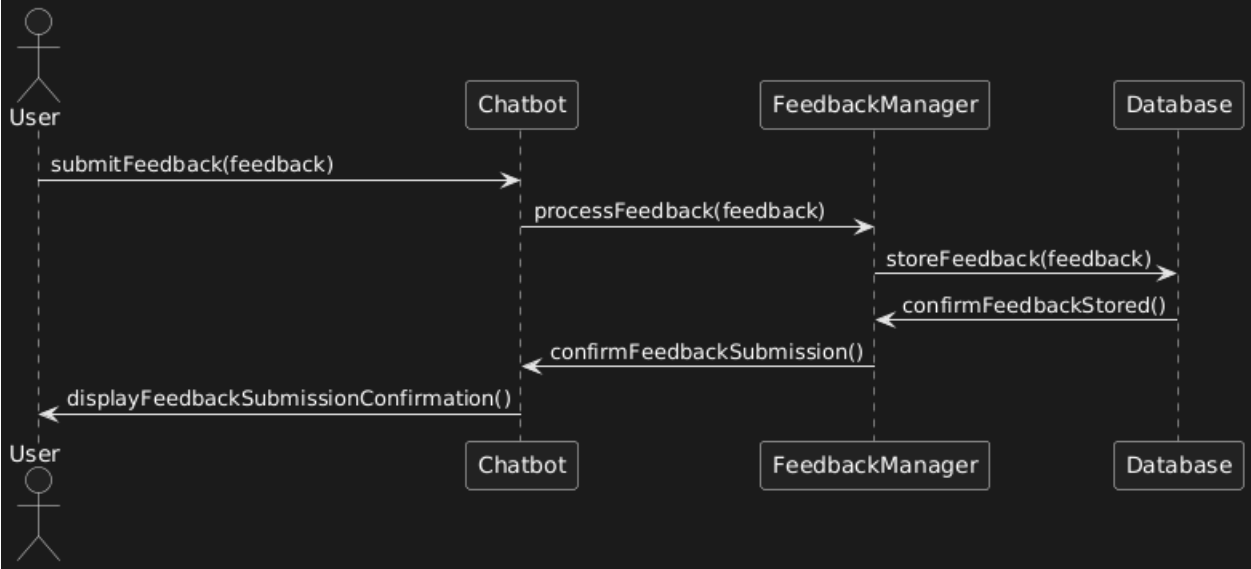
Skin Detection:



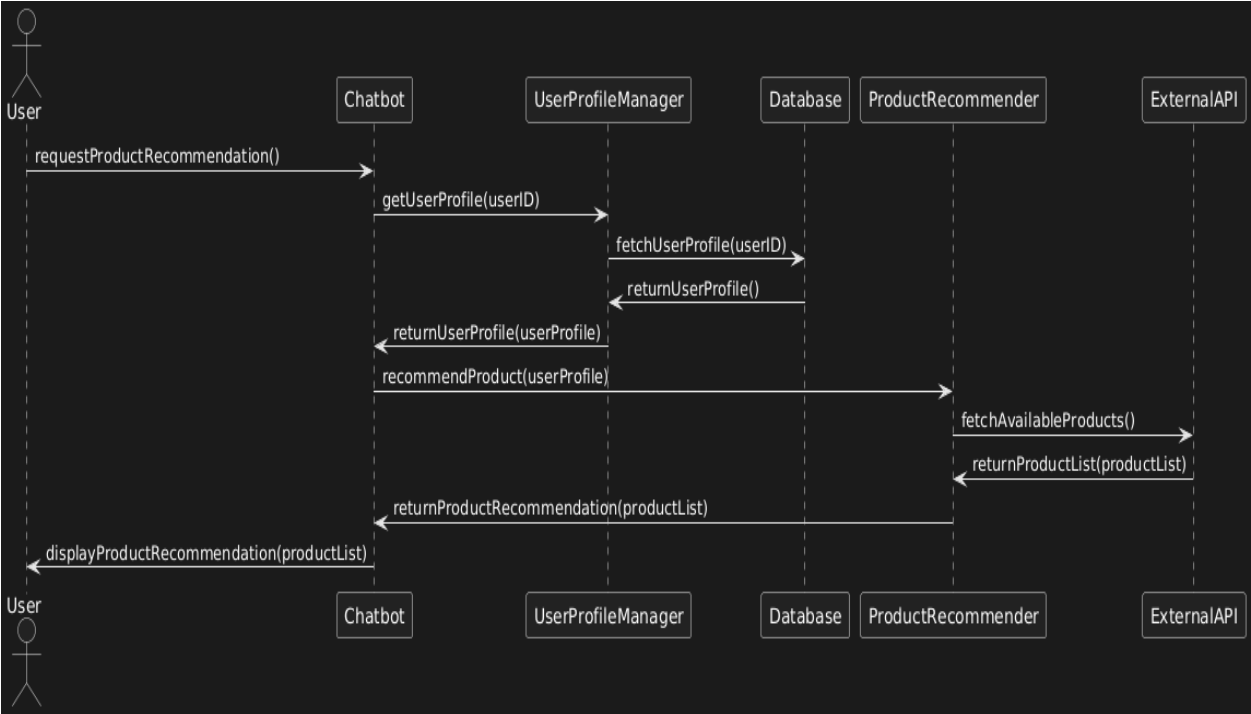
Diet recommendation:



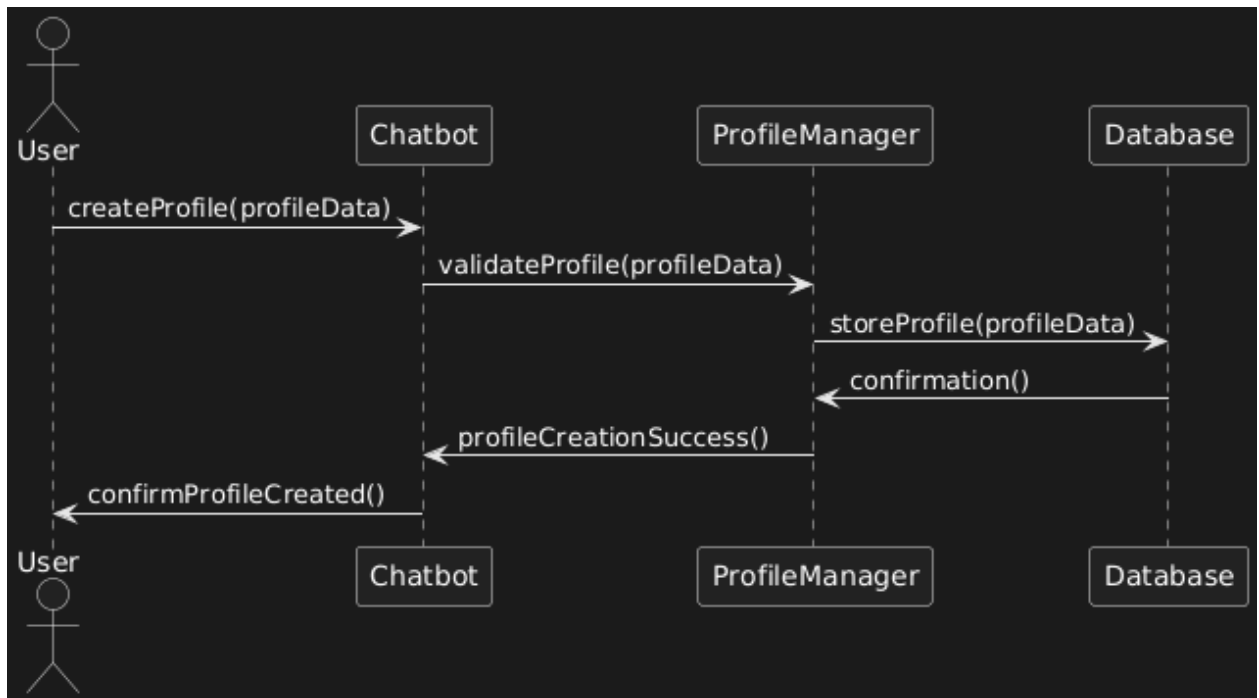
User Feedback:



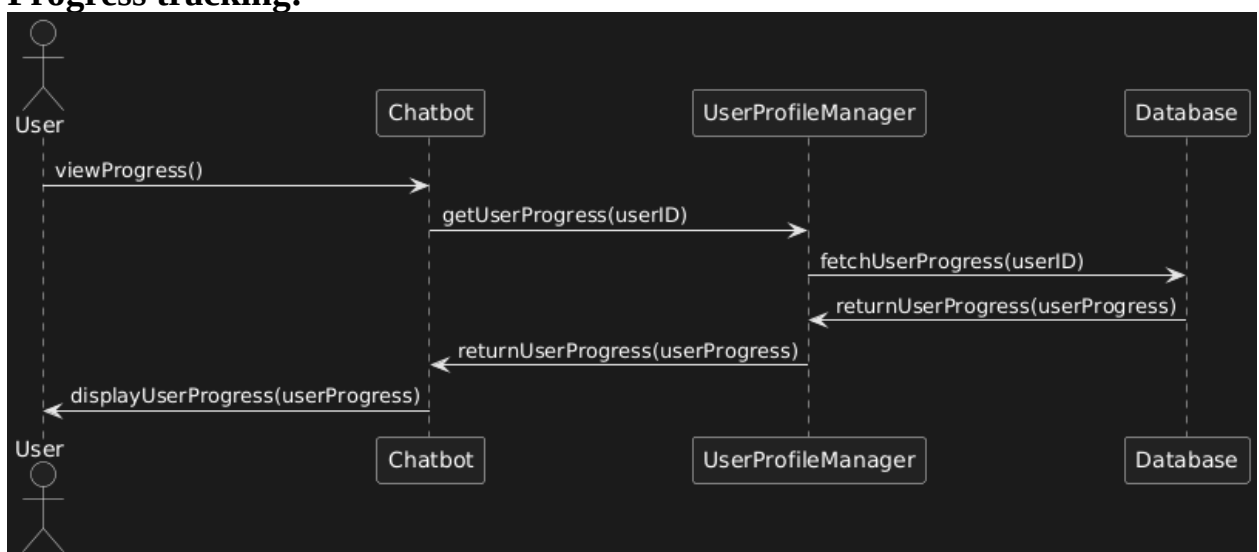
Product recommendation:



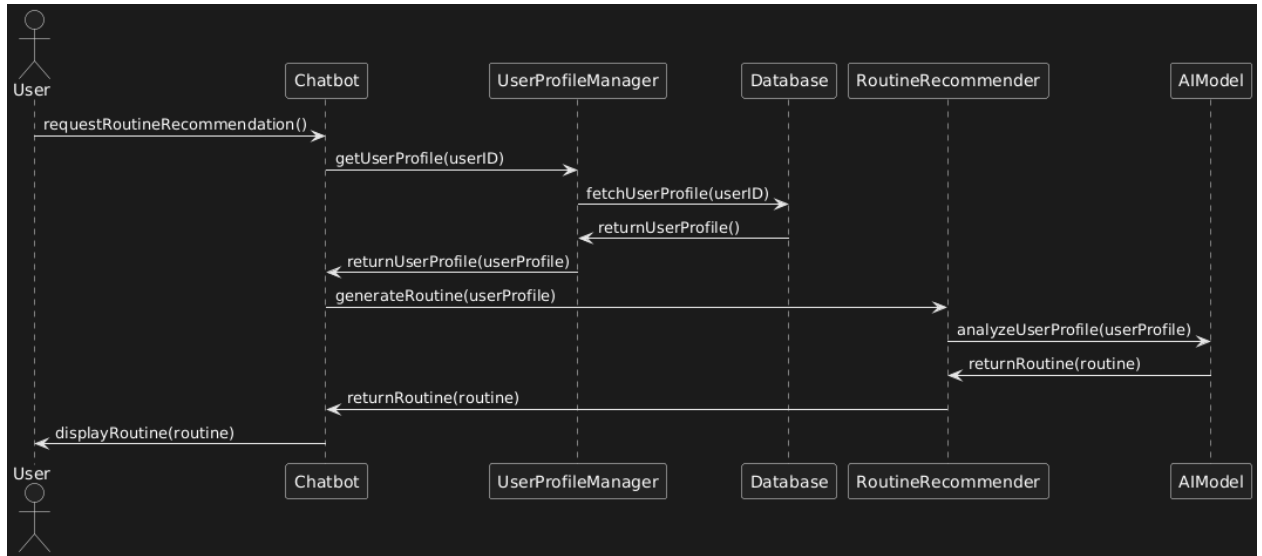
Profile Creation:



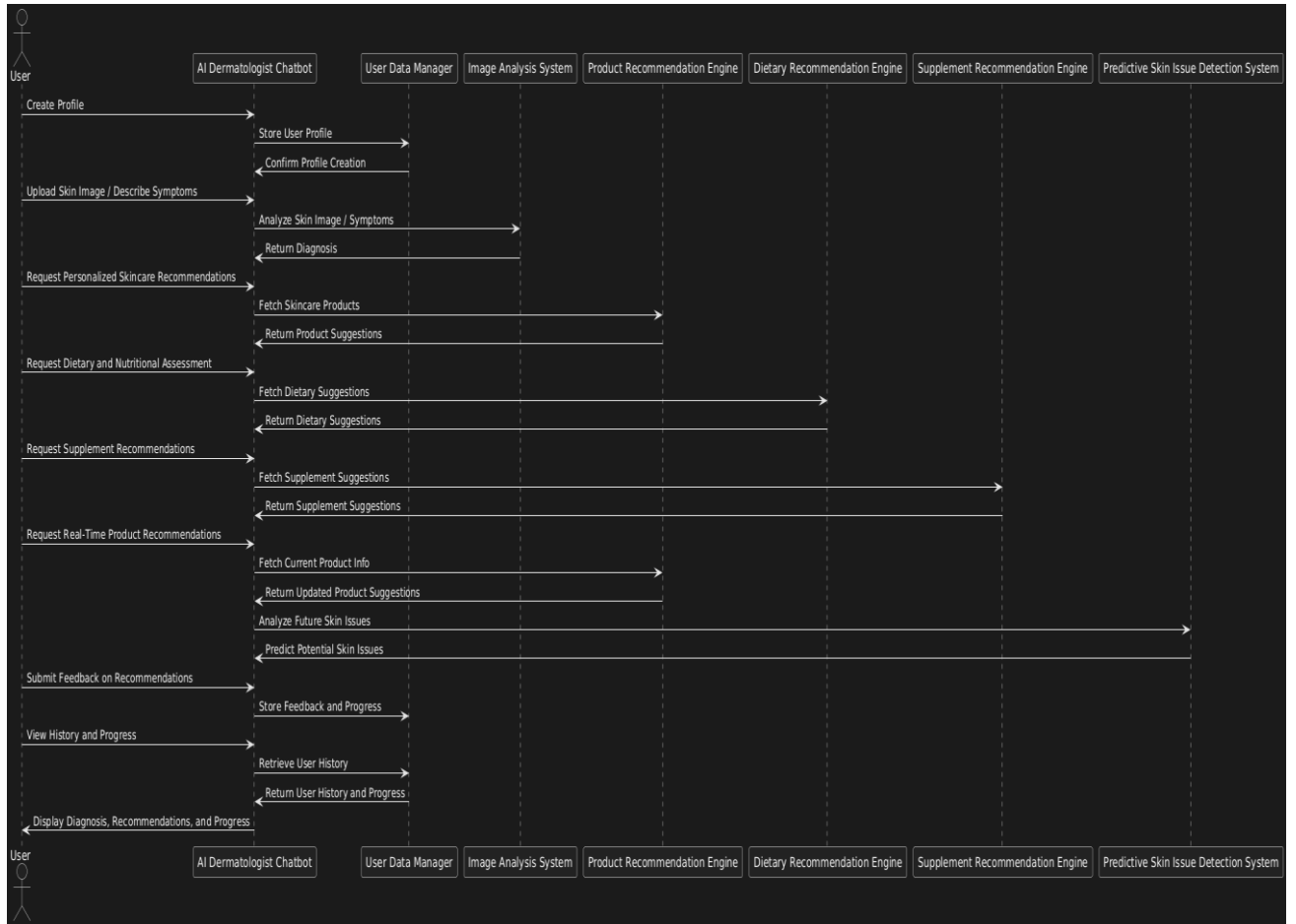
Progress tracking:



Routine Recommendation:



System Level Sequence Diagram



Software Design

Design Methodology: Object-Oriented Programming (OOP)

The Object-Oriented Programming design is well-suited for the development of the system due to several key factors which I will discuss below :

Modular Design and Reusability:

Firstly, OOP makes it possible to separate the system into various classes and give each with a distinct function. For instance, separate classes can be built for components such as DietaryRecommender, ProductRecommender, ImageAnalyzer, and Chatbot, each of which is responsible of carrying their own tasks within the class and system

Object-oriented programming makes it easier for us to scale the system. By either new class can be created, or existing ones which could be modified with little to no code modifications to add new functionalities into our system.

Enhanced Maintainability:

Better maintainability is provided by object-oriented programming, which makes sure that every object or class controls its own data and functions and communicates with other parts via designated interfaces. This showcases that modifications that are made to a single system component can be made without influencing any other component in the system.

Encapsulation:

OOP encapsulation makes it possible to isolate core states and behavior especially those managing user data, image analysis, or recommendation system that are within each class. This design promotes safe, as well as regulated interactions between various system components they do this by guaranteeing that all inner workings of one class remain concealed from other classes present.

(b) Architectural Style: Microservices Architecture

For the AI Dermatologist Chatbot, adopting a Microservices Architecture is considered the most suitable architectural approach. The reasons listed below are why we decided to use this approach:

Independent Services:

The system may be divided into multiple components, including the Image Analysis, Product Recommendation Engine, Dietary Recommendation Engine, and User Data Manager, because of the microservices architecture. This will help us when we want each service to be developed, launched, and grown on its own.

This split is advantageous to our project since the skin analysis, user feedback, user profile management, and recommendation services may all function independently before we integrate them. For instance, improving the image analysis process is made feasible without actually jeopardizing the stability of other system elements.

Scalability and Flexibility:

Microservices enable independent scaling. If the skin analysis feature is experiencing heavy usage, only that part of the system needs to be scaled. The other components, like the product recommendation service, can remain unaffected.

Fault Tolerance:

In a microservices architecture, the failure of one service doesn't crash the entire system. For example, if the product recommendation service experiences an issue, other critical services such as user profile management and feedback submission will continue operating seamlessly.

Simplified Deployment:

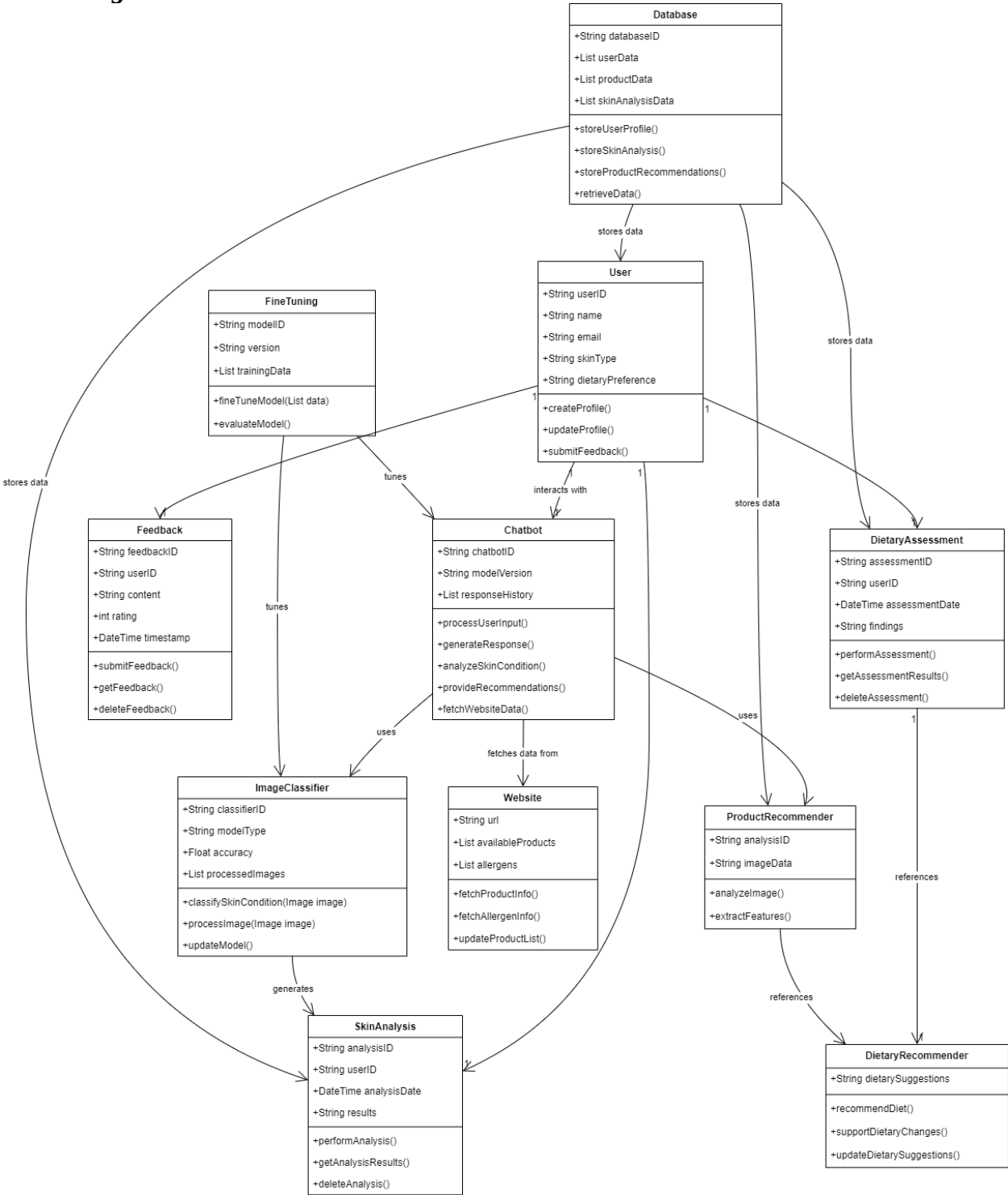
Workflows including continuous integration and continuous deployment (CI/CD) are best suited for microservices. Updates don't always need to affect the entire system; they might be applied to specific services. This enables your system to react to user requests and new features with greater agility and responsiveness.

Technology-Agnostic Development:

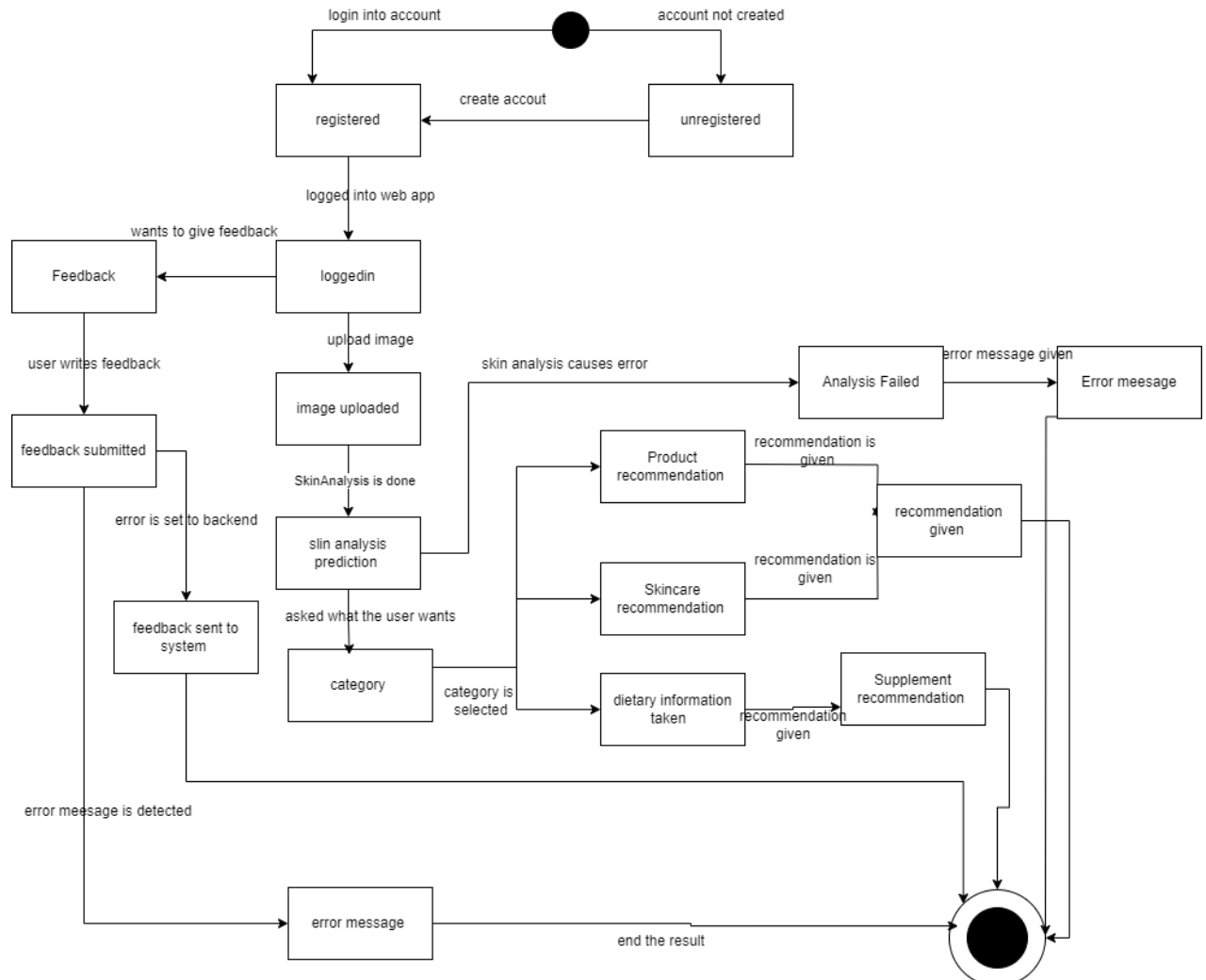
Every microservice can be constructed using the technology that best fits its needs. For instance, you might rely on a separate stack for the chatbot interface and utilize Python and machine learning frameworks for the analysis of skin picture data. This adaptability makes it possible to find more ideal solutions for every system component.

Behavior Design (Design Phase)

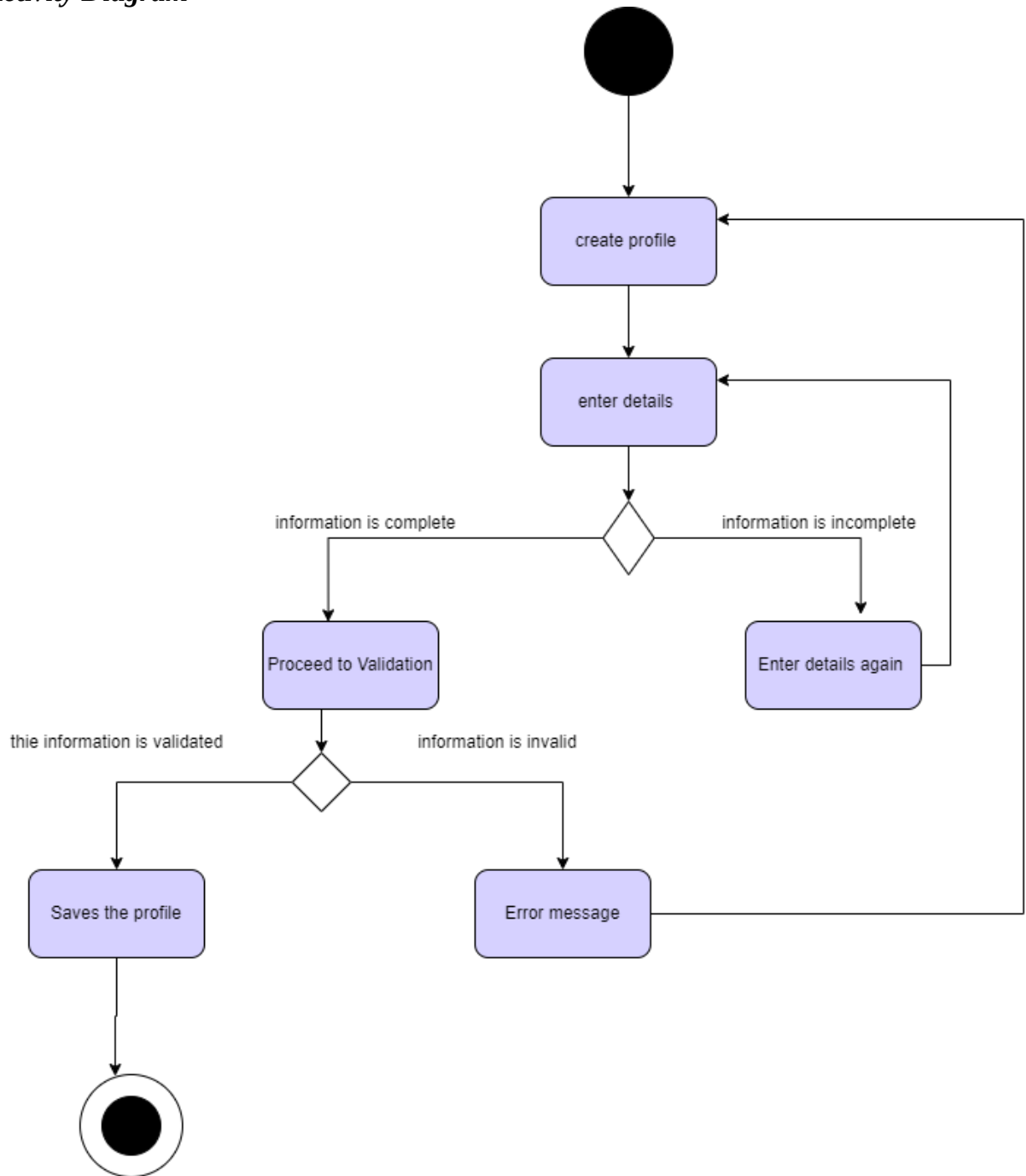
Class Diagram:



State Transition Graph:

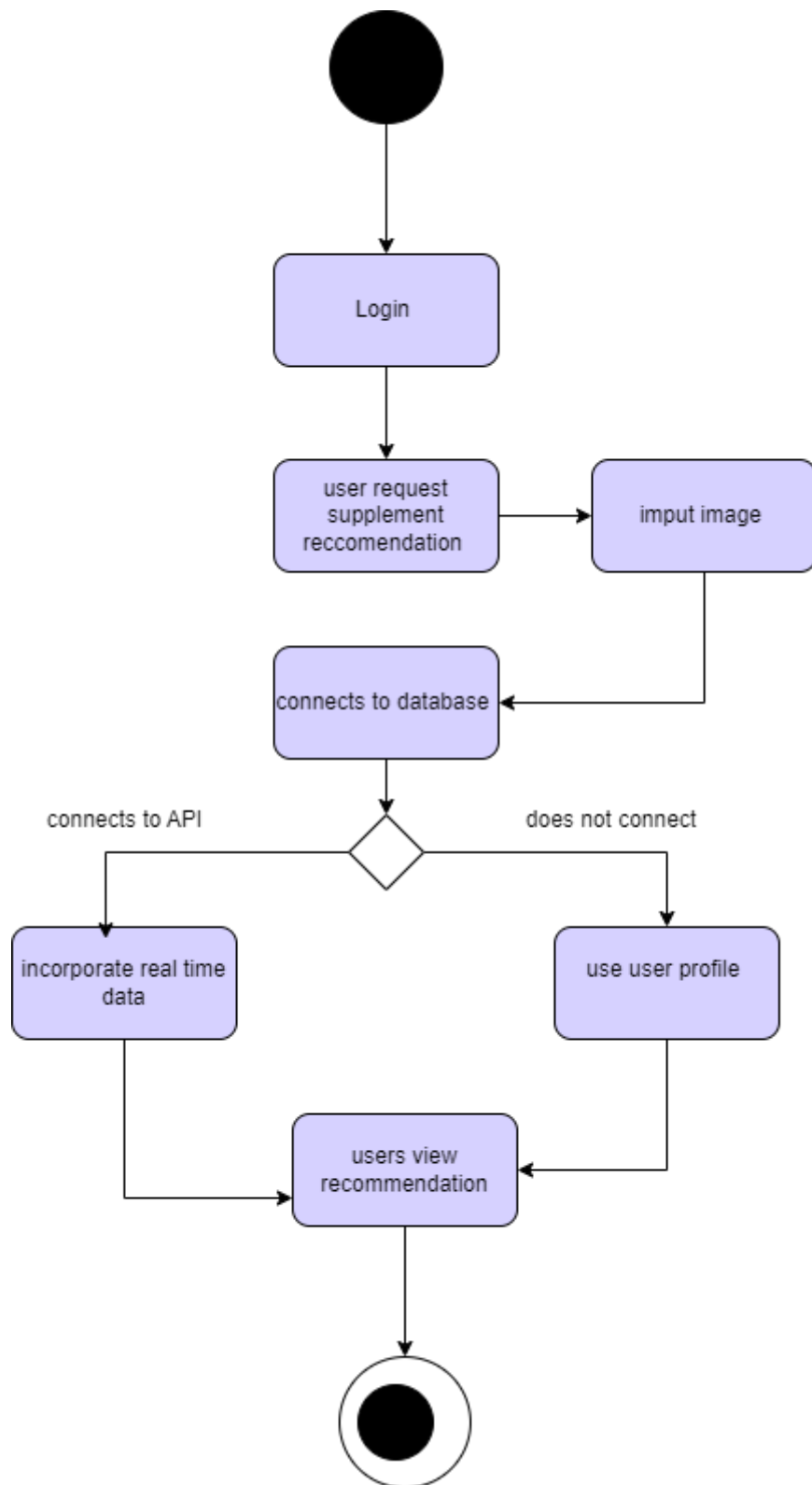


Activity Diagram

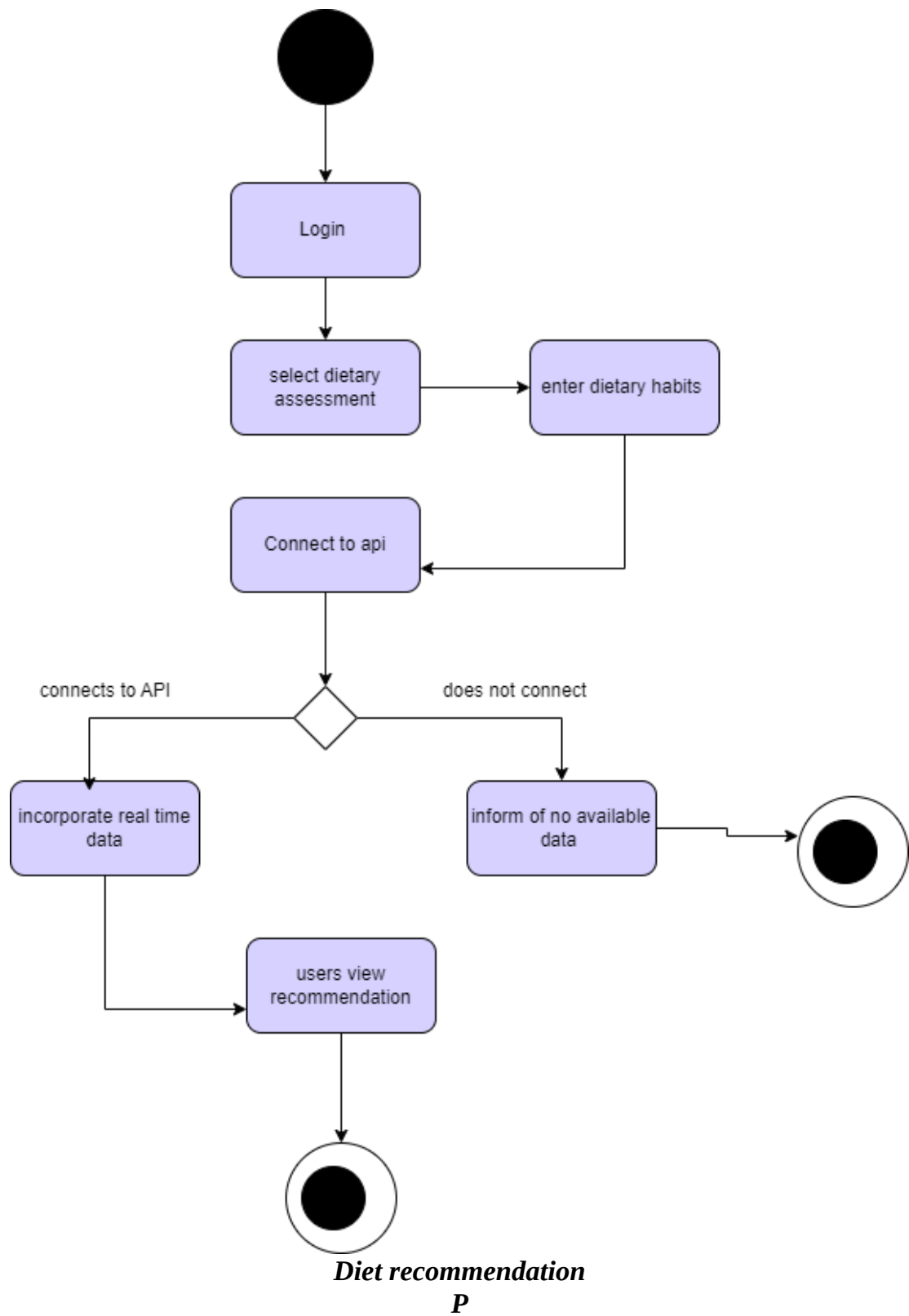


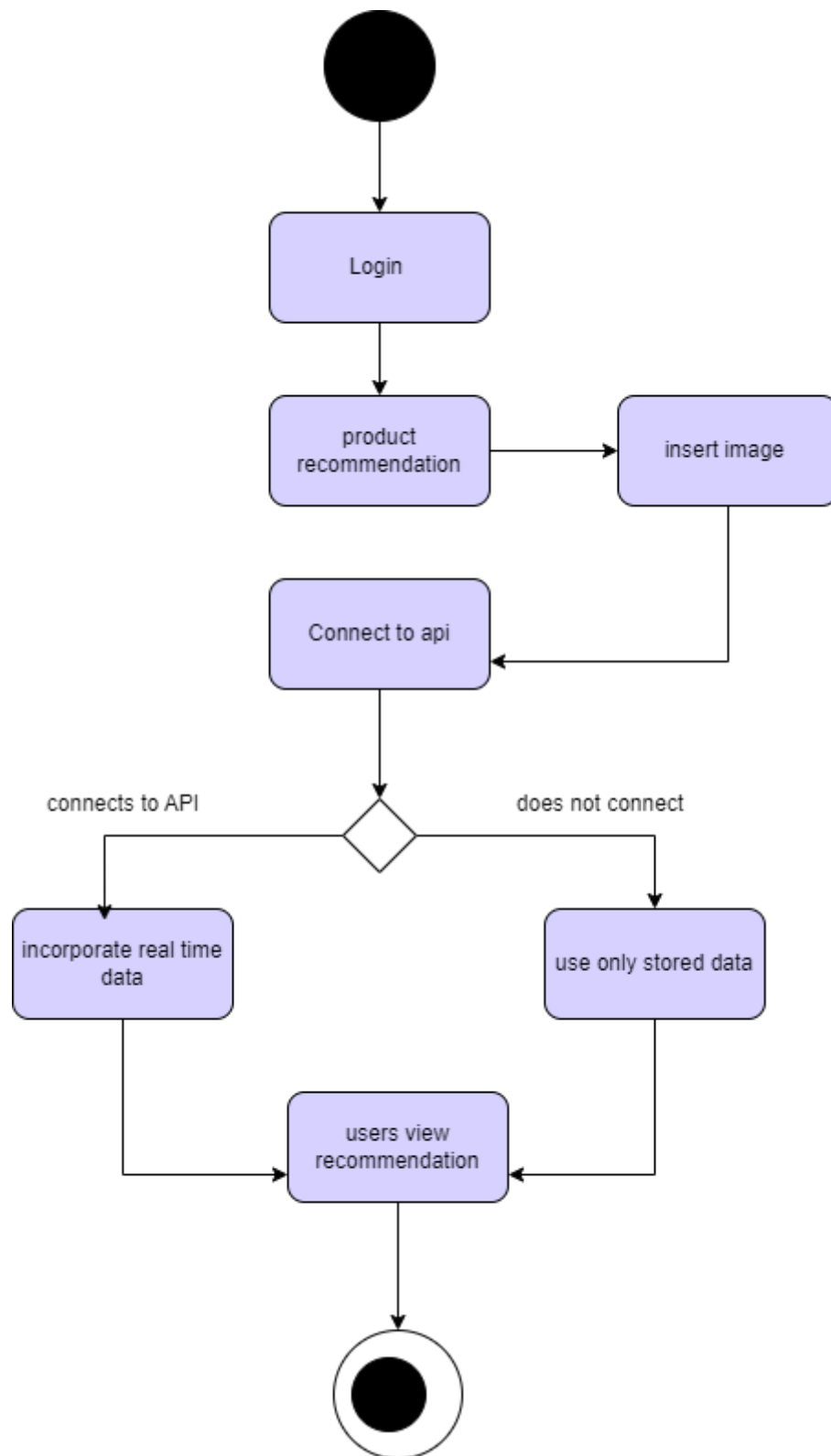
1.Profile creation

2.

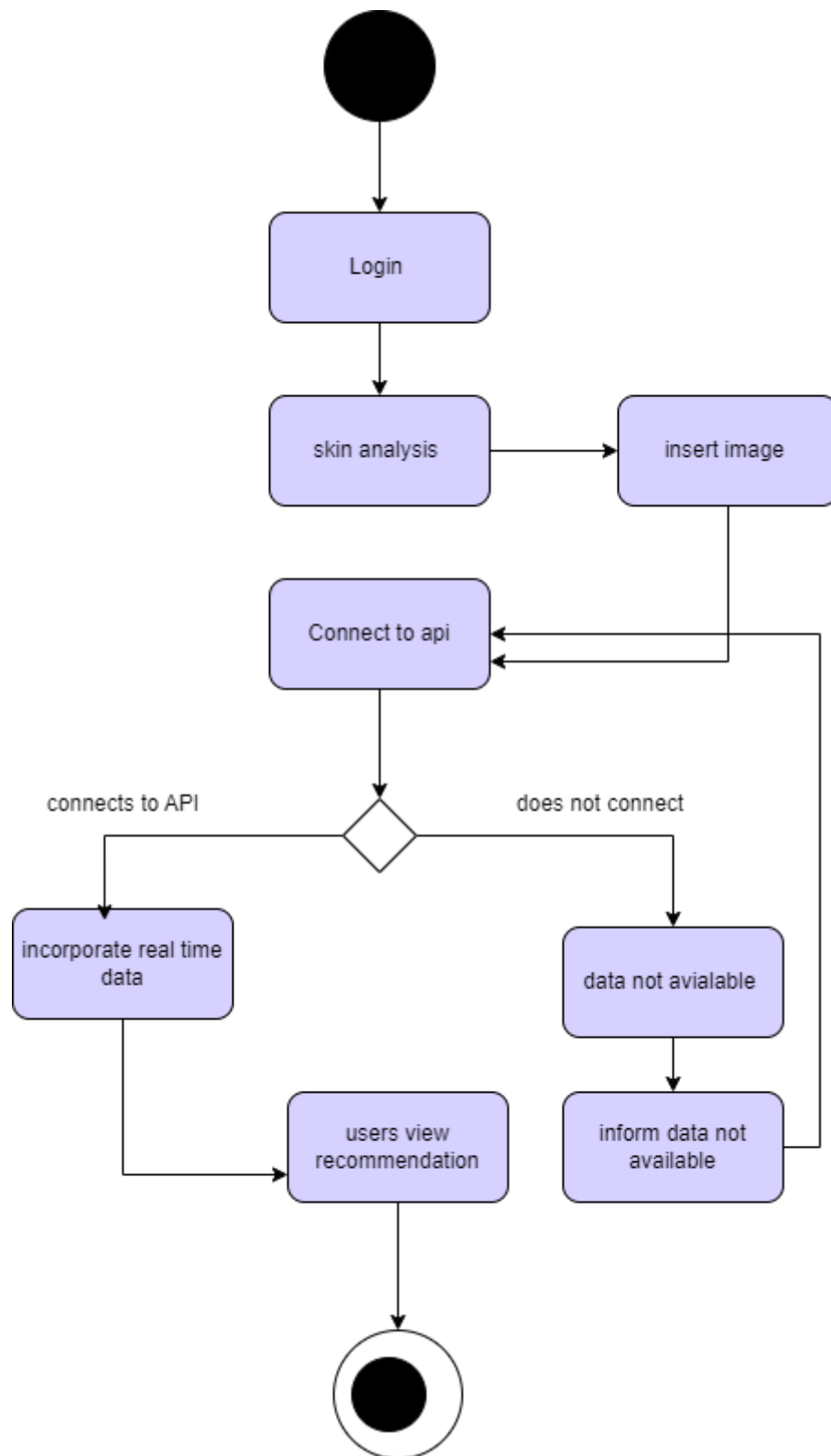


Supplement recommendation

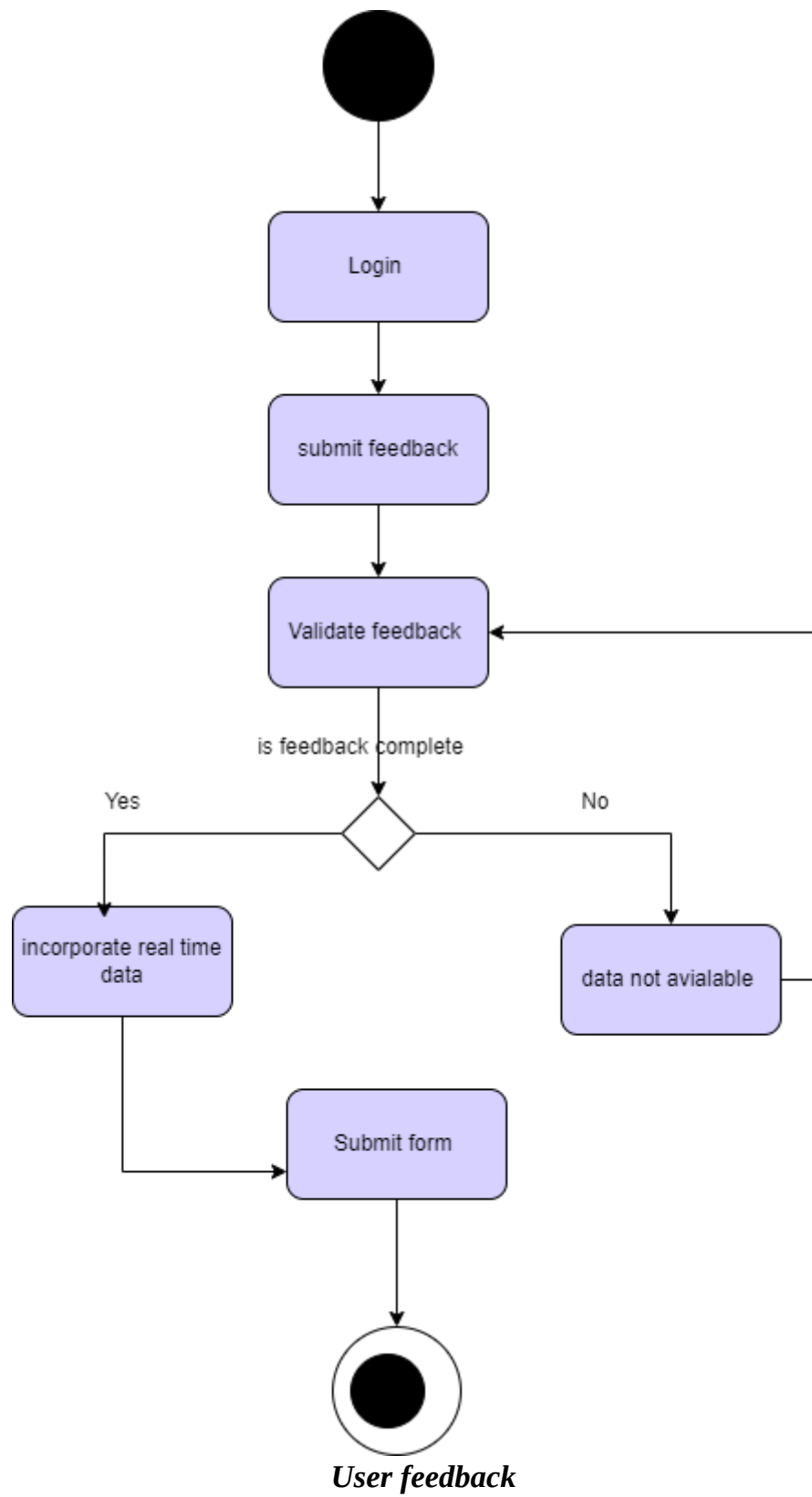


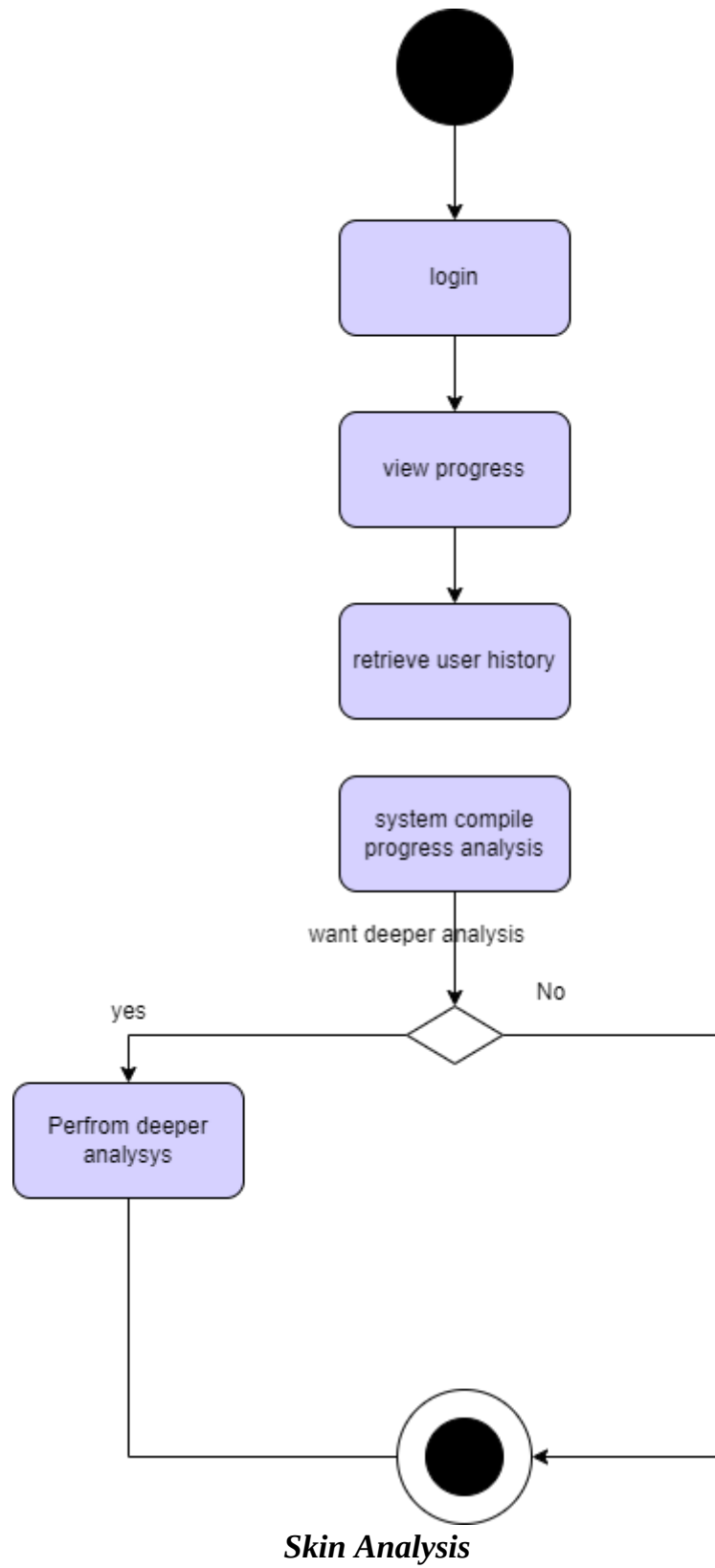


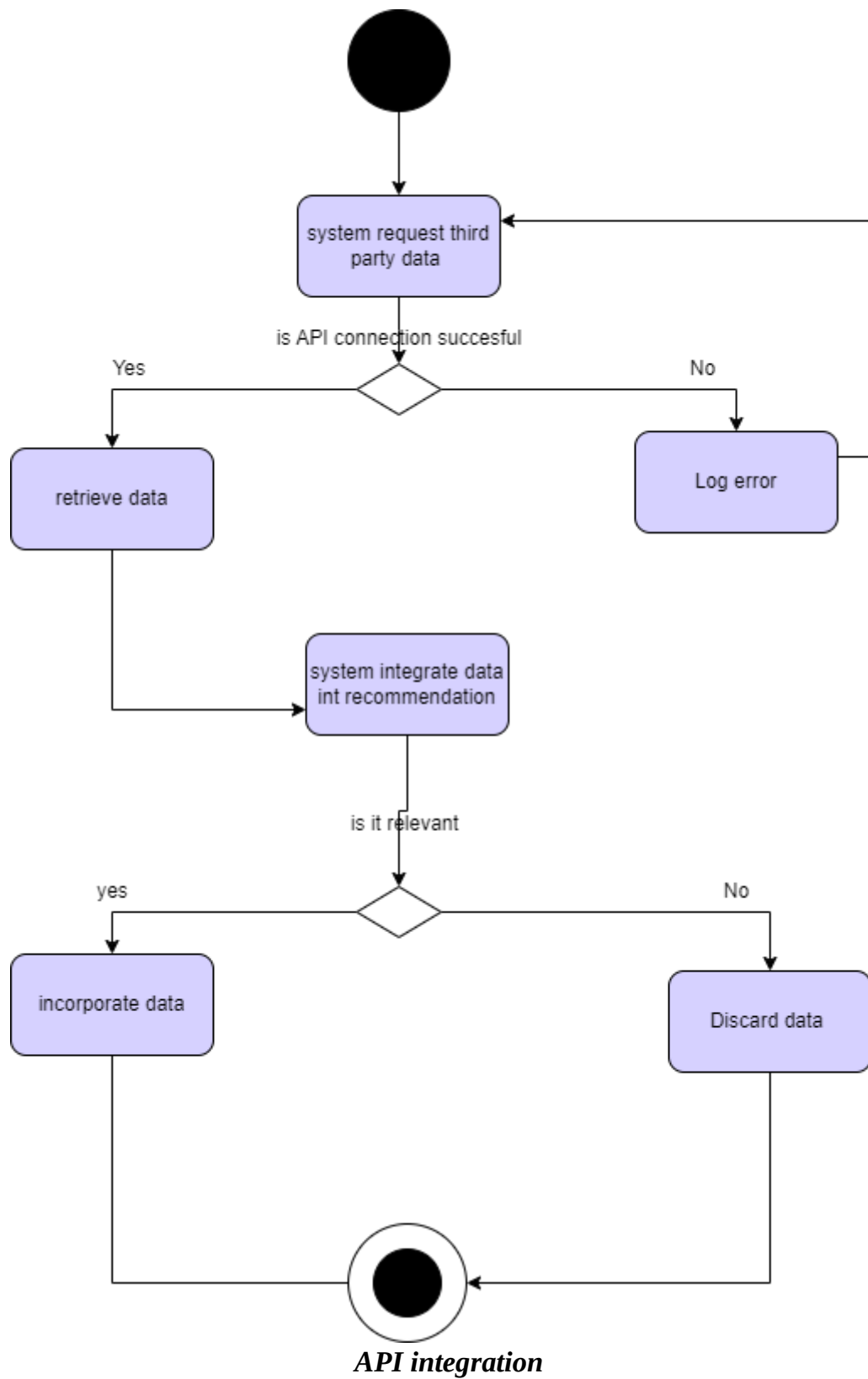
Product recommendation



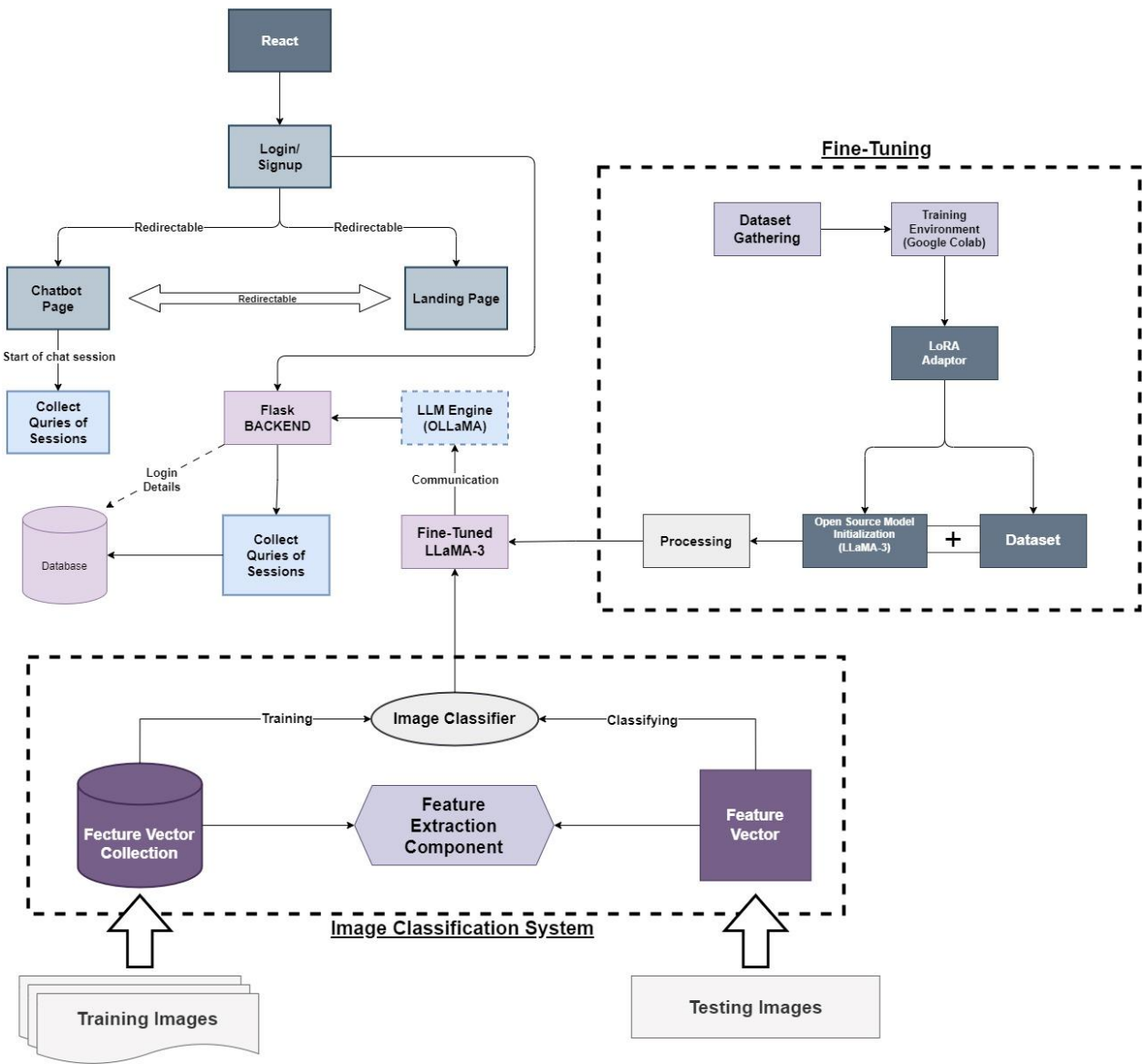
Skincare recommendation

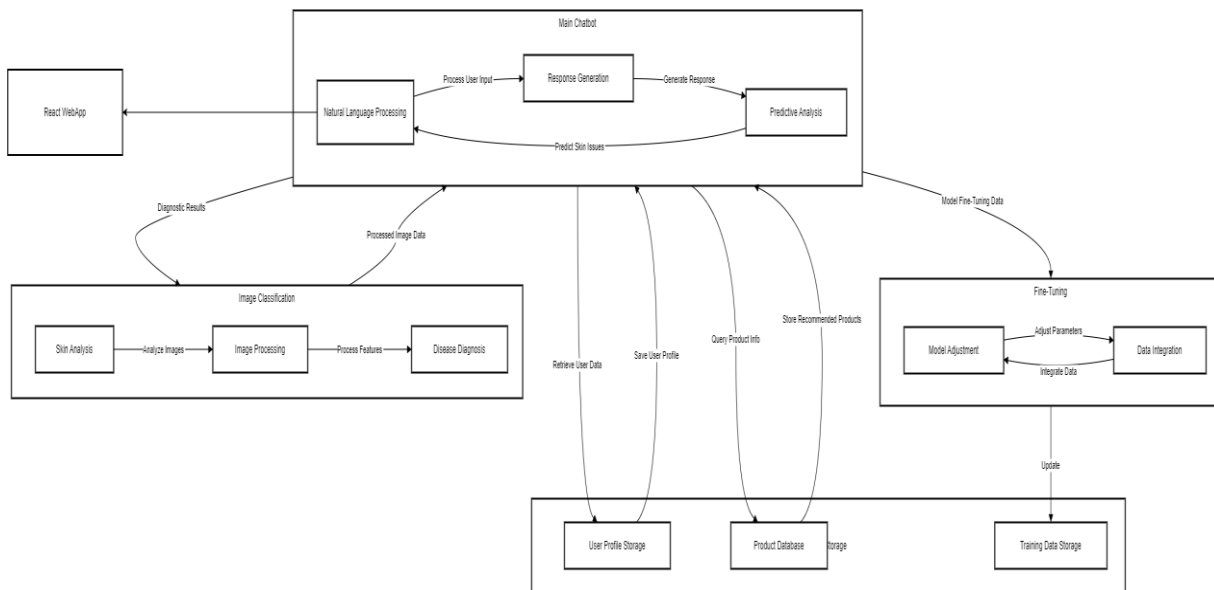




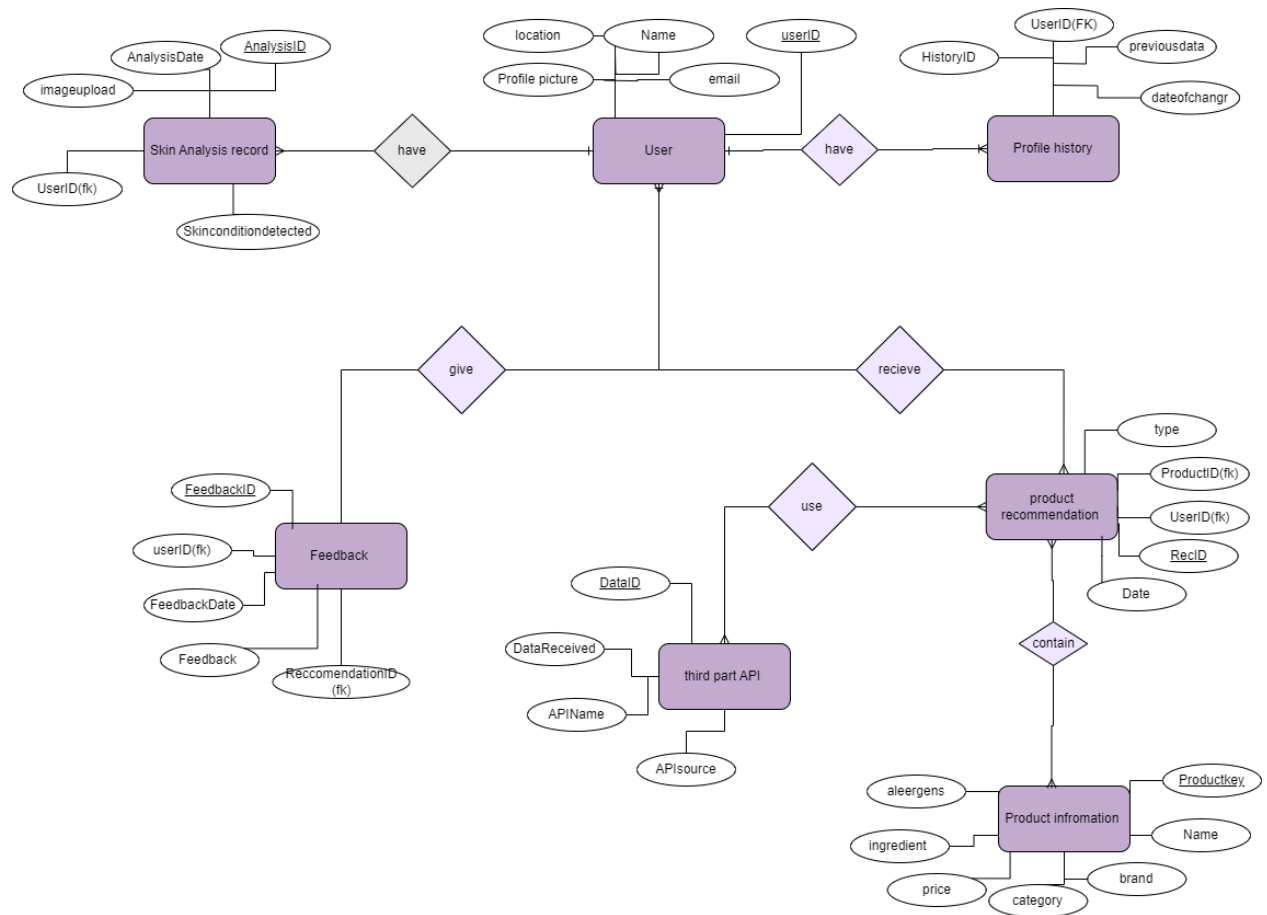


Architectural Diagram:





ER Diagram (DB Representation):



Conclusion:

This report examines the whole project overview and provides a detailed analysis of how we will be taking forward this project for our FYP. We have provided all the details from idea research, competitive analysis, problem statement implementation design, iteration plan and development. We have provided comprehensive diagrams to understand the behavior of our project. We have complied with all the SE protocols and produced this document. The document and all its components are our intellectual property.

References:

Used APA Citation for references:

- Paluri, K. V., & Gupta, A. (2024). ETLoViT: An acne diagnose approach using vision-transformers and model ensembling. *Engineering Research Express*, 6(3), 035242.
<https://doi.org/10.1088/2631-8695/ad7ad9>
- Ghadekar, P., Joshi, A., Vanjari, A., Raza, M., Gupta, S., & Gajaralwar, A. (2022). *Acne detection using convolutional neural networks and image processing technique*. In D. R. Panigrahi, & P. K. Mallick (Eds.), *Advances in Manufacturing and Industrial Engineering* (pp. 47-54). CRC Press.
- Arief Kelik Nugroho, Retantyo Wardoyo, Moh Edi Wibowo, & Hardyanto Soebono. (2024). Image dermoscopy skin lesion classification using deep learning method: systematic literature review. *Bulletin of Electrical Engineering and Informatics*, 13(2), 1042–1049. <https://doi.org/10.11591/eei.v13i2.6077>
- Jones, S., & Chan, H. (2023). The Psychological Effects of Skin Disorders. *Journal of Dermatological Science*, 12(3), 56-67.

